FLIP, FREEZE, FLOW: EXPLORING INTERACTIVE SONIFICATION OF BREAKDANCING

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ABSTRACT

Whilst movement sonification has been applied within sports training, motor rehabilitation, and several modern dance forms, it has not yet been researched in the context of the dance style of breaking (breakdancing in common parlance) despite significant potential to enhance dancer expressivity and audience engagement. As part of a student project, we designed and developed a real-time camera-based sonification system able to respond to various aspects of the dancer's movement in a musical and intuitively movement-congruent manner using a sample-based audio architecture in Max/MSP. Parameter mappings were grounded in six conceptual movement-sound interactions informed by the fundamental movement vocabulary specific to breaking. Through real-life tests and interviews with two professional dancers, we found that the movement-sound interactions based on vertical motion (e.g. head height on the Y-axis) were the most intuitive and engaging, although the unintended activation of certain other effects caused frustration. In addition, the perceptually subtle mappings were often overlooked or ignored, highlighting the importance of salient sonic changes to the dancer experience. Nevertheless, we believe that our technological framework and mapping design philosophy can inform future research on the sonification of breaking.

1. INTRODUCTION

Breaking, a highly dynamic street dance originating from the African-American and Puerto Rican communities in New York, is typically set to drum breaks and sample-based funk, soul, and hip-hop music [1]. Characterized by intricate footwork, explosive power moves, freezes and other signature moves, breaking combines rhythmic precision with athleticism and improvisation. This paper uses the term *breaking* rather than "breakdancing", as the former is the original term used by practitioners, whereas the latter is a term coined by popular media[1]. Breaking has steadily grown in popularity over the years [2] and has gained global recognition, culminating in its inclusion in the 2024 Olympic Games [3]. Despite its cultural significance and increasing presence in mainstream stage productions (e.g. the Danish dance company Uppercut Dance Theatre[2]), research on integrating breaking with emerging technologies, particularly sonification, remains scarce. With the advent of low-cost motion sensing tools and the growing prevalence of high-performance computing, it has become increasingly possible to augment various dance forms through real-time multimodal interaction to enrich performer and audience experiences alike. The exploration of breaking sonification has never been more culturally relevant [4] or technically feasible, which serves as the primary motivation behind the present work.

Movement sonification translates movement variables (e.g. speed, position, etc.) into sound [5, 6], typically by mapping data from inertial sensors or camera-based motion-tracking systems to real-time audio synthesis or processing parameters[7]. It is applied in many domains ranging from sports training [8] and movement rehabilitation [9, 10] to art exhibitions [11], dance performances [12, 13] etc., and may play a variety of roles in these contexts ranging from artistic to therapeutic. In essence, movement sonification mediates motor perception and performance by adding sound to otherwise silent movements, with the precise movement-sound mapping dependent on the exact role of the sonification as well as the capabilities/preferences of the user [7, 8, 9, 14]. In the realm of dance, real-time movement sonification can serve as a tool to not only enhance learning through enriched sensory feedback on motion [15] but also support artistic expression [4, 12, 13].

Despite the expanding body of dance sonification research at large, there exists (to the best of our knowledge) little to no work specifically focused on breaking, an especially challenging dance form to sonify due to its combination of complex and high-energy movements, full-body involvement, and improvisatory nature. This gap represents an underexplored, yet golden opportunity to explore new modes of artistic expression and interaction between dancers and sound. Unlike contemporary dance, which is typically designed for the stage and readily amenable to systematic experimental approaches to performance [16] (like Merce Cunninghams approach[3]) the raw and individualistic nature of breaking demands alternative approaches and a more adaptable design philosophy.

Taking inspiration from the dynamic conversation and relation between drummers and dancers in many African dance cultures[17, 18], this work aimed to serve as an initial exploration of what might happen if *the music reacts to the dancer* as opposed to the more traditional reverse. The broad design goal was to sonify breaking in a manner that feels engaging and intuitive to the

[1] RedBull: This is why we say breaking instead of breakdancing

[2] Link to the history of Uppercut Dance Theatre

[3] Link to about Merce Cunningham

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dancer, an experience which would, at the surface level, necessitate a sonification capable of spontaneously reflecting the evolving energetic and positional dynamics central to the art of breaking. Through this research, we hope to contribute to the broader discussion on how movement sonification can be meaningfully applied to culturally rich and physically demanding dance forms such as breaking.

2. RELATED WORK

2.1. Breaking and its Movement Vocabulary

Breaking, also known as *b-boying* or *b-girling*, emerged as a dynamic and expressive dance form within hip-hop culture. It originated in the 1970's in the Bronx, New York, influenced by African American and Latin youth who used dance as a form of self-expression and competition [19]. DJ Kool Herc pioneered the technique of extending instrumental breaks at block parties, allowing b-boys and b-girls (the dancers) to showcase their most acrobatic moves. This innovation played a key role in shaping hip-hop dance and music culture[20, 21]. The music accompanying breaking usually makes use of high-energy rhythms, syncopated beats, and repetitive percussive loops. Traditionally there is a focus on groove and rhythm to support the improvisational nature of the dance. DJs manipulate records to highlight these instrumental breaks, creating a musical canvas for dancer expression. Over time, producers have expanded the sonic palette of breaking music, incorporating electronic elements and digital sampling techniques while maintaining its emphasis on rhythm and movement [22]. Although a range of breaking techniques exist, dance movements in breaking can be very fast, powerful, explosive and dynamic [23], appearing almost to defy gravity in some instances. At its core, breaking can be said to be made up of the following primary movement gestalts [4] [5]: *toprock* (standing steps), *downrock* (ground-based footwork), *power moves* (acrobatic spins and flips), and *freezes* (held poses that punctuate movements).

2.2. Dance Sonification

Movement sonification has been tested in numerous contemporary dance contexts over the years, with novel interactive paradigms becoming increasingly possible with the advent of low-cost wearable sensors, advancing camera technology, and modern machine learning for pattern recognition. Research on dance sonification employs various mapping metaphors. The "body as an instrument" metaphor directly maps dancers' movements to sound parameters, treating gestures as musical inputs. "Spatialized sound" links movement across a stage to perceived sonic placement, immersing the performer in an interactive sound environment. These are some metaphors we have come across in literature.

Several key studies have shaped the field, providing insights into different sonification strategies. Merce Cunninghams *Variation V* (1965) is one of the earliest examples of dance sonification, using antennas, and later, photoelectric cells to capture the contemporary dancers movements and trigger sounds. The approach set an example of interactive movement-sound systems, demonstrating the potential for movement-activated music. The sound was influenced by both the dancers' movements and the control of musicians, creating a unique relationship between dance and sound. *Variation V*[6] revealed the flexibility in collaboration between dancer and musician, as the musician could not fully predict the dancers movements. Ventura Dance Company's *Heliopolis* (2014)[7] is the most recent performance with a combined camera-based motion tracking with gesture recognition software to generate music dynamically, creating an interactive relationship between the dancers and the performance environment. This project demonstrated the potential of real-time movement sonification in professional contemporary dance performances, and was praised for its audiovisual interaction, creating a multi-sensory immersive experience, as the dancers movements also influenced a video projection. *Otterbein et al* studied movement sonification in contemporary dance[24], using wearable sensor bands with force-sensing resistors (FSRs) and inertial measurement units (IMUs). They explored expressive mappings using both low-level motion descriptors (musle contraction, acceleration) and high-level Laban Movement Analysis descriptors (weight, time, space, and flow), where FSRs were linked to muscle contraction and release, and Laban effort descriptors were mapped to visual and sound features (Granular synthesis and particle-based visuals). While the system allowed rich movement exploration, complex mappings was difficult to control intuitively. *SonifyFOLK* [25] (2023) is a web-based application for sonifying movements in dance performances, applied and evaluated with Swedish folk dance. It uses an optical infrared motion capture system with full-body markers on dancers and a fiddle player, tracked via MoCapToolbox for MATLAB. Designed for intuitive use, it provides accessible parameter mapping for non-experts, so the dancers can create their own mappings in the interface. The study highlighted the importance of immediate visual and acoustic feedback when adjusting parameters. *Y. Theologitis* [26] (2024) emphasizes that effective sonification of contemporary dance relies on intuitive movement-sound mapping, real-time auditory feedback, and expressive parameter control, where motion speed and intensity dynamically shape pitch, volume, and timbre, enhancing both performer experience and audience perception through MediaPipe and IMU-based motion capture. *Sounding Bodies* (2024) explores how sonification of physical contact affects behaviour in contact improvisation dance. Using the TouchMe device, two sonifications (pleasant vs. tense) were tested. Results showed that sonification influenced interaction, with pleasant sonification leading to more contact, implying that sonic design choices can directly influence how dancers interact with a system.

These studies and projects illustrate different design philosophies, from direct motion-triggered sound generation to more nuanced mappings that enhance expressive potential. Building on these approaches, our project sonifies breaking using camera-based movement tracking, specifically employing MediaPipe to translate movement features into sound parameters [4]. Additionally, movement will influence sound placement in space, creating an immersive environment, like in Variation V and Heliopolis. The sonification will enhance both performer interaction and audience experience, as learned in the studies SonifyFOLK and Sounding Bodies. For breaking, a key challenge is preserving the dances visual and kinetic intensity while ensuring meaningful sonification of its energy, complexity, and rhythmic nuances. By integrating metaphorical approaches from prior research, this project explores how breakings acrobatic, explosive, and rhythmic characteristics can be effectively translated into sound.

[4] Red Bull article breaking down the basic elements of breaking

[5] Video showcasing breaking elements and personal expressions.

[6] https://www.eai.org/titles/variations-v

[7] https://www.ventura-dance.com/works/heliopolis

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2.3. Framing the Present Work

This paper addresses the development of a dance sonification system specifically for breaking. Our research question was framed as follows: *"How can a real-time breaking sonification system be designed to enhance expressivity, accommodate the full range of bodily movements, and provide intuitive auditory feedback, while being easy to understand?"* At the outset, we framed the following basic requirements for the system:

- The mappings must be specifically tailored to the movement vocabulary of breaking.
- The movement-driven sonic changes must be perceived to be movement-congruent, responsive, and reliably triggered.
- The mappings must enhance expressivity without restricting the movement of the dancer.
- The mappings should be easily understandable to a dancer.

3. DESIGN

3.1. Sound Design

The music follows a break-beat structure, inspired by J Dilla's[8] signature rhythmic style [27]. All loops were created using MIDI and VST instruments within Cubase and Ableton. The drum and percussion arrangement begins with a simple drum beat, using open hi-hats to mimic the style of classic hiphop samples. The second and third layers added syncopation using short percussive sounds, by placing the notes and accents to complement the open spaces left in the first loop. The rhythmic layers were chosen on the basis that breaking has three primary levels where the dance operates that were obvious to exploit: toprock, downrock and power moves. Harmonic stabs, influenced by disco and R'n'B samples, served to add punctuation, with string and guitar stabs assigned to the left and right hand, respectively. The bass loops were chosen to reflect classic hip-hop grooves from the 1980s and 1990s, accentuating the first and third beats of the bars, while incorporating minimal yet syncopated riffs to maintain groove and rhythmic variation.

3.2. Movement-Sound Interactions

The system mappings were designed so as to accommodate five discrete movement-sound interaction (MSI) ideas (link to demo: *M1-M5*) based on breaking-specific movement vocabulary and movement-sound associations:

- **Energy of the dance (M1)**: When dancers transition from dancing upright to being closer to the ground, the movements tend to include fast spins and intricate footwork. These movements with higher physical intensity typically accompany denser and more complex drum rhythms in breakbeats. Hence in this interaction, when the dancer gets lower to the ground, the drum patterns become more layered and complex.
- **Vacuum (M2)**: High-risk acrobatic moves such as handstands, power moves or flips, heighten audience anticipation of the dancer landing the move. This is reflected in breaking music, where DJs or producers sometimes momentarily cut the music or add a filter to enhance and build suspense. For this interaction, we designed a "vacuum" effect meant to sound as if the music itself holds its breath.
- **Freeze (M3)**: Freezes, the moments when the dancer suddenly holds a position, often balancing in an intricate position on one hand, their head or combinations, are signature characteristics of breaking. In this interaction, when the head is on the floor, a beat repeat effect is activated, creating a visual and rhythmic punctuation.
- **Rooting (M4)**: Rooting in breaking refers to establishing a strong connection to the beat through foundational movements. In this interaction, the bass line serves as a musical anchor, remaining simple when the dancer is centered, symbolizing stability, and becoming more complex when moving towards the sides, representing dynamic expression.
- **Stabs (M5)**: Sharp, isolated gestures, such as fast arm movements, kicks or entering a freeze, serve as rhythmic accents within breaking. These are often mirrored in DJing, where scratch techniques or percussive accents ("stabs") are used to emphasize hits in the music. In this interaction, fast arm movements trigger chord stabs.

4. IMPLEMENTATION

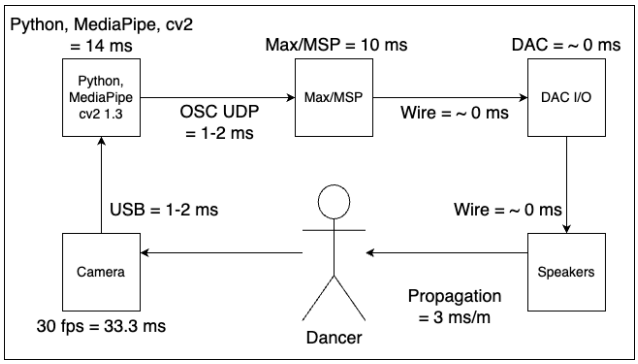


Figure 1: Block diagram of the implementation, highlighting the data flow from the dancer to sensing, audio output, and back to movement. Estimated latency values are indicated alongside each block.

Our system employed camera-based sensing, real-time pose recognition through MediaPipe, and sound generation using Max/MSP by Cycling 74. The sensing was chosen for its ease of implementation and access to positional information of all body parts without the need for hardware. Max/MSP was selected for its versatility, real-time operation, visual interface, and easy availability of documentation. Each component contributes to the overall system latency as shown in Fig. 1.

4.1. Real-Time Movement Tracking

To ensure flexible placement, we chose the Logitech C920 HD PRO web camera as the sensing device over a built-in webcam. To extract real-time skeletal positional data from the raw camera feed, we used MediaPipe's '*Pose Landmark Detection*' solution[9] for its

[8] Review of the book "Dilla Time"

[9] https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker

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accuracy, real-time operation, lack of required external hardware, and interoperability. We implemented a Python script to extract bodily landmarks of interest and transmit their 3D positional data to Max/MSP as *Open Sound Control (OSC)* messages.

Chosen Landmarks: By default, the Pose Landmark Detection model identifies and tracks 31 keypoints (landmarks) pertaining to the human skeleton from a plain camera feed. We selected a subset of seven landmarks numbered as per MediaPipe convention: nose (landmark 0), wrists (landmarks 15 and 16), hips (landmarks 23 and 24), and ankles (landmarks 27 and 28). Choosing only seven landmarks representative of the entire body both improved computational performance and reduced transmission latency over OSC while capturing the essential components of the breaking vocabulary. The *nose* landmark was chosen to represent the head. The *hips* served as a stable reference point, crucial for balance, weight distribution, and movement coordination, providing a robust anchor point for tracking overall posture. The *ankles* represented the distance to the floor, useful in capturing kicks, jumps, or the upside-down movements that make up some of the more complex breaking vocabulary elements.

Model Configuration: MediaPipe provides 3-D spatial coordinates for each selected landmark along with a reliability score that indicates the model's confidence level in the detected position. In terms of Python libraries, our implementation uses *cv2* for video processing, *Mediapipe* for pose detection, and *udp_client* from *pythonosc* to transmit data to Max/MSP. The MediaPipe pose recognition model was initialized with '*model_complexity=0*' to minimize computational load (least complex model out of three). An OSC client was created to transmit selected landmarks ([0, 15, 16, 23, 24, 27, 28]). The camera, accessed via *VideoCapture(0)*, was set to 320x240 resolution to reduce processing demands. Frames were read, flipped for a mirror-like view, and converted from BGR to RGB. The pose recognition model ran with *'min_detection_confidence=0.5'* and *'min_tracking_confidence=0.5'*, to ensure reliable tracking. If detected, landmark coordinates (rounded to four decimals) were updated, with wrist values smoothed for stability. Data was sent via OSC, visualized with *imshow*, and the program could be terminated with 'q'. Upon exit, the camera feed was released, and OpenCV windows closed to free resources.

OSC Communication: The system tracks a set of pose landmarks ([0, 15, 16, 23, 24, 27, 28] - refer Table 1), and sends their spatial coordinates via OSC. For each detected landmark, four separate messages are sent, corresponding to the X, Y, and Z coordinates, as well as the visibility score. The messages has the following format and order: `x`, `y`, `z`, and `visibility`. Messages are sent at the cameras frame rate, which is 30 FPS. In optimal conditions, this results in 28 messages per frame and up to 840 OSC messages per second.

4.2. Max/MSP

Max/MSP receives OSC information from MediaPipe, which is then divided into landmark-specific floating point data that are scaled to empirically set bounds on a parameter-specific basis prior to mapping.

Interaction-specific Mapping: The technical realization of each MSI (link to demo: *M1-M5*) is summarized in Table 1, and Fig. 2 depicts the zones pertaining to M1, M3, and M4.

Nose - Vertical Position - M1 - As the dancer's head moves lower, the system maps the nose's Y-axis position (scaled 0-100, bottom to top of frame) determines loop complexity, increasing as the nose moves lower. The frame is divided into three height regions, with each lower zone adding a new percussive layer. This increases rhythmic complexity to go with dancers' typically intensifying movement dynamics the closer they get to the ground.

Ankles - Delay and Filtering - M2 - As the dancer becomes airborne or the ankles are getting high, the system sonifies this by mapping the ankles Y-axis positions to control two parameters: increasing delay mix and decreasing high-pass filter frequency as the feet rise, simulating a vacuum effect. Each ankle independently influences the effect, with the full effect applied when both ankles reach maximum height. Values are smoothed to reduce jitter, and a low-pass filter is applied to the delay feedback for a softer sound. Scaling is logarithmic for a naturalistic perceptual scaling.

Nose - Beat Repeat - M3 - When the nose reaches a predefined lowest threshold, a beat repeat effect is applied to the drum section. This effect, commonly used in hip-hop and breaking, activates only on quarter notes and repeats every 16th note while the nose remains in the activation zone.

Hips - Bass Loop Triggers - M4 - The system maps hip position along the X-axis to control bass loop complexity. When the dancer stays within a central range, a simple bass loop plays. On moving outside this range (left or right), a more complex layer is added. This allows the dancer's lateral movement to dynamically shape the bassline.

Wrists - Stabs - M5 - The system sonifies sharp, fast wrist movements by tracking speed along X- and Y-axis. The values (scaled 0500 frame-edge to frame-edge) are used to calculate wrist speed. When the latter exceeds an empirically set threshold, a gain control is briefly raised, creating a stab sound effect. Each wrist independently triggers a different stab sound to add variation.

Hips - Panning - Azimuth Position - Stereo panning is linearly controlled by mapping the scaled X position of the hips to the balance between left and right mono signals.

Visibility - Master Fader Control - All visibility values are summed, and if the total is zero (i.e., no body parts detected), the master fader lowers output to silence and triggers a vinyl backspin sound effect.

4.3. Final Mixdown

The systems multitrack audio output is mixed down to stereo, with individual track gains set through referencing over different headphones and loudspeakers to create a balanced soundscape aligned with modern breaking mixes. Except for the "visibility" sound effect, all sounds pass through a master gain slider and a Max/MSP compressor prior to final output.

5. EVALUATION

For the practical evaluation of the system, two professional dancers (26 and 33 years, with respectively 20 and 19 years of dancing experience) were selected. Both had extensive experience with breaking, and were involved with dance companies specializing in incorporating breaking into stage performances. The evaluation took place at a performance research lab, approximately 6m $\times$ 6m (test layout in Figure 3). The setup comprised a Logitech C920 HD PRO webcam, a MacBook Pro 16" (2019, 2.6 GHz) laptop, a Focusrite Clarett+ 4Pre audio interface, and KRK Rokit 5 speakers. The webcam was placed 2550 mm from the front of the dance

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Name	Movement-Sound Interaction (MSI)	Landmarks	Axis	Parameters
M1	Drum pattern complexity increased through layer addition as head goes lower down	# 0 Nose	Y	Gate threshold and Volume gain
M2	Delay mix raised and and high-pass frequency lowered as the ankle(s) increases in height	# 27 & 28 Ankles	Y	Low-pass filter cutoff and delay mix
M3	Beat repeat triggered as head height enters lowest zone (corresponding to headstand)	# 0 Nose	Y	Beat repeat activation
M4	Bass loop complexity increased through layer addition when dancer moves towards either side	# 23 & 24 Hips	X	Bass loop 2 activation
M5	Chord stabs triggered as wrist speeds cross a predefined threshold	# 15 & 16 Wrists	X,Y	Volume gain with envelope
Azimuth Position	Stereo pan position of full mix follows the lateral position of the dancer within the frame	# 23 & 24 Hips	X	Balance - left and right gain

Table 1: Assignment table depicting how the five conceptual MSI's were realized in terms of data-sound mappings.

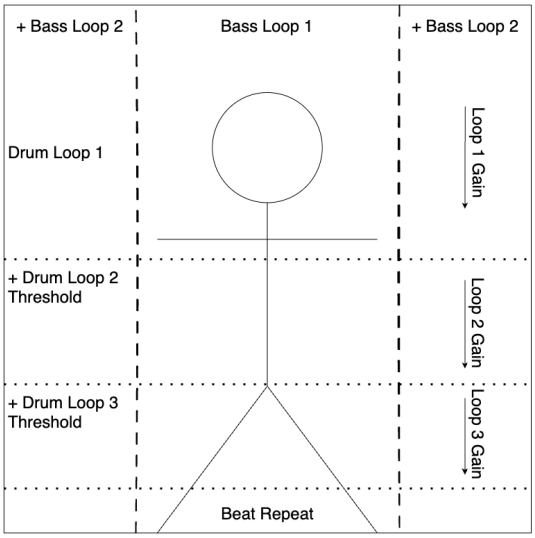


Figure 2: Movement-sound mapping based on the dancer's position. The **vertical axis** controls drum loop complexity: as the dancer (nose tracking) moves downward, additional drum loops are layered. Each dotted horizontal line represents a threshold where a new loop is introduced, and its volume increases as the dancer continues to move lower. The lowest zone activates a beat repeat effect. The **horizontal axis** controls bass loop variations. Moving within the center zone, a simple bass loop plays, while shifting left or right, introduces an additional layer, making the bass loop more complex.

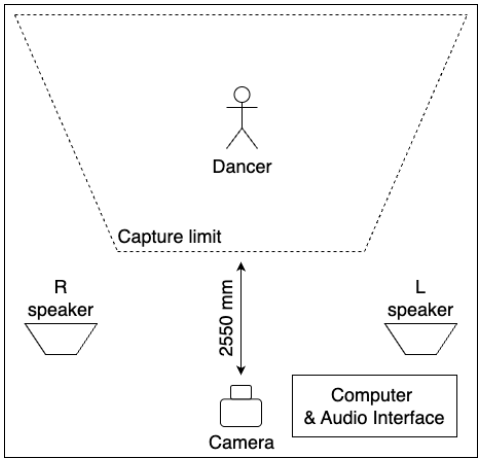


Figure 3: Top view of test layout (not to scale).

area and 225 mm off the ground, oriented so that the front edge of the dance area was at the frame bottom.

The test involved each dancer freely interacting with the system over two rounds and then participating in a structured interview. Observations were noted, and audio recordings of the system output were also captured to enable subsequent analysis of how the dancers interacted with the system. A Focusrite Scarlett 2i2 audio interface was used to record the audio output stream, and the triggering / activation of the various sounds was tracked as MIDI events to facilitate visualization. Both spontaneous feedback and structured interviews were recorded, transcribed, and analyzed to identify patterns in the dancers' experiences and approaches. The analysis was twofold, consisting of a high-level behavioral analysis of the recorded audio and a qualitative summarization of the dancers' experiences and suggestions.

5.1. Results

5.1.1. Behavioral Analysis Based on Audio Data

The participants varied in terms of their interaction patterns. Figure 4 illustrate when P2 triggered different beat zones and Figure 5 illustrate all of P1's gestures. P1 appeared to engage less with vertical zones (M1 and M3), while P2 triggered all beat zones more frequently (seen as longer notes in the blue and yellow horizontal blocks on Fig. 5 and shorter notes in all three horizontal blocks on Fig. 4). P1 activated M3 (beat repeat) and M2 (delay and filter) while having their head against the floor, followed by acrobatic movements (seen as black dots in the red (M2) and dark green (M3) block accompanying the long notes in the yellow block on Fig. 5). Both participants maintained similar horizontal movement, as reflected in steady panning and bass activation (M4)(exemplified by P1, seen as steady spread of black notes in the brown, orange and muted red blocks in Fig. 5) The dancers

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appeared to do a lot of fast arm movements with M5 (exemplified by P1, seen as many notes in the turquoise and coral blocks on Fig. 5). P2 appears to have taken a more structured approach, evenly distributing triggers between the first two zones, indicating frequent transitions. In contrast, P1 spent longer in individual zones but used more arm gestures to trigger guitar and string stabs. In summary, P2 relied more on vertical movements for beat changes, while P1 relied more on arm gestures to trigger the stabs.

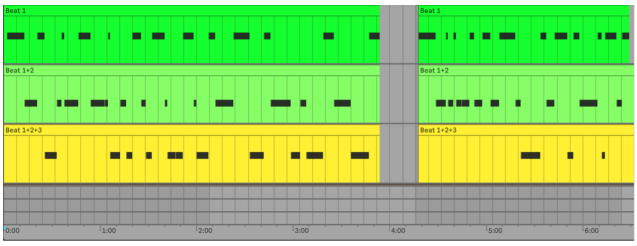


Figure 4: P2, beat zones triggered in time, and time spent in individual zones (length of notes). The block on the top represents zone 1 with beat loop 1, the middle zone 2 with beat loop 1+2 and the bottom zone 3 with beat loop 1+2+3.

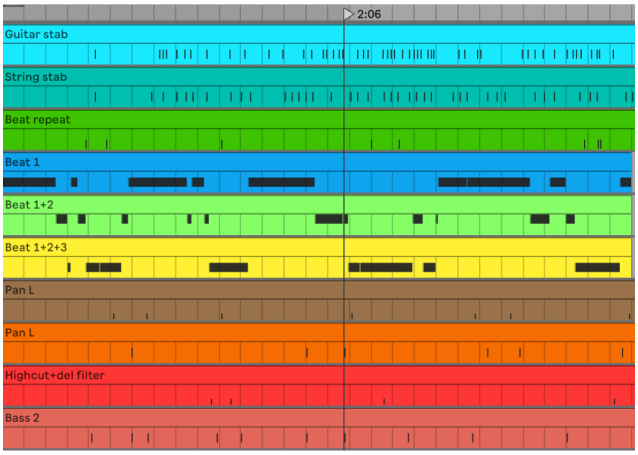


Figure 5: Gesture mapping of P1 dancing, run 1. The notes (black boxes, short and long, represent the interactions with the different mappings of the system. Light-blue and coral: stabs(M5). Dark green: beat repeat(M3). Blue, green and yellow: beat zones(M1). Brown and orange: panning. Red: filter + delay (M2). Muted red (lowest block): complex bass loop (M4). For instance, at 2:06, P1 was dancing in a squat position on the left side of the frame, triggering bass loop 2, while having just triggered several guitar stabs.

5.1.2. *Interview Analysis*

Overall, P1 and P2 both saw potential for live performance applications and liked the idea of using the body as a musical instrument, shaping the music through movements. However, they pointed out that dancing to unfamiliar beats made it harder to immerse themselves. In general, they suggested a more personalized experience, such as being able to use their favorite song, smoother beat transitions, and having more interesting and easy-to-trigger feedback when lifting their legs. Expanding the sonic palette with extra layers, such as vocal loops, was also suggested to maintain engagement over longer performances. Finally, they highlighted the importance of incorporating spin-tracking into the system, as spinning is a key movement in breaking. In terms of interaction-specific feedback:

- *Lagging (M1):* Both agreed that the beats getting busier and more complex as the head lowered was a very desirable effect, influencing them to move faster when low and slower when upright. Returning to a standing position at the center of the frame allowed them to "rest" as the beat simplified. However, they occasionally felt the system lagged behind their movements in time.
- *Improved foot feedback and sensivity (M2):* They suggested refining foot mappings to provide clearer feedback and adjusting foot sensitivity to ensure more responsive interactions.
- *More responsive beat repeat (M3):* Their experiences with the beat repeat differed. P1 found its delay interesting and playable, while P2 preferred it to be snappier and more responsive, struggling with the activation and release times.
- *Issues with Stabs (M5):* Both dancers found the stabs unnatural in the context of breaking, as they were short, repetitive, and often triggered unintentionally by incidental arm movements, for example when used for balance explained by P1. P1 suggested that varying the stab sounds could introduce an element of surprise and making hand gestures more engaging to compensate for the lack of precision.

Notably, neither dancer initially noticed the stereo panning, M2, or M4 until these were explicitly pointed out between the runs. From both analyses, it is evident that the Y-axis MSI's were the preferred. The first two zones of the beat levels had the highest level of intentional interaction. Although triggered often by the dancers, the wrist stabs were found to be distracting and often unintended, resulting in frustration and self-consciousness. The more subtle X-axis MSI's worked as intended but their lack of perceptual salience by the dancers led to a lack of exploration of these specific parameters. The beat repeat was used as often as the X-axis MSI's, and from the interviews, its slow responsiveness was flagged as an issue. From the audio data, the M2 was the least explored overall; as these were arguably the hardest to activate through movement, this was not altogether surprising. However, the audio data and interview data reveal a discrepancy: unlike the wrist stabs, which were frequently triggered unintentionally, the intended activation of certain effects was often unsuccessful, leading to a mismatch between the auditory output and the dancers' movements.

6. GENERAL DISCUSSION

In this work, we designed and built a real-time breakdancing sonification system based on camera tracking and Max/MSP, and subsequently conducted an exploratory real-life test with two dancers. The tests highlighted how the learning curve was minimal for both participants, who were confidently using it by their second trial and, broadly speaking, finding the experience to be enjoyable, game-like, and fun. The system was generally deemed responsive, with the M1 MSI receiving the most positive feedback and engagement. Gesture data indicated that the two upper percussion zones were preferred over the lowest zone. M2 needs refinement,

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as the effect was only properly audible at extreme foot elevation, which often is not achieved in real-life breaking. Additionally, responsiveness varied with dancer height, thus a need for better scaling and threshold adjustment. The arm gestures were the least positively received, primarily due to their overly sensitive nature. Gesture data confirmed this, by the decreased usage in the second trial. This was most likely due to patch inconsistencies, indicating the need for further refinement of the motion data processing and threshold settings.

Our findings align with prior research on movement sonification. We found that vertical motion was particular effective for intuitive sonification, similar to the work of Otterbein *et al.* [24] who observed low-level motion descriptors are well-suited for real-time feedback. However, our work uniquely addresses breaking's high-energy, acrobatic nature, emphasizing mappings reflecting its nuances. Our results also resonate with SonifyFOLK [25], which emphasized the importance of immediate acoustic feedback. Breaking's fast movements demands faster temporal precision. A novel aspect of our work was that we introduced breaking-specific MSI's, such as the "vacuum" effect. These MSI's were inspired by the cultural and musical context, specifically the role of DJ's. Our findings suggest that basing sonification in cultural nuances of the dance enhance both performer engagement and audience experience. Finally, challenges in balancing sensitivity an precision, particularly with wrist-triggered stabs, reflect Theologitis work [26], highlighting the need for careful calibration in fast-paced dance forms like breaking. Despite these issues, the system maintained a high degree of reliability, with stable tracking and zero crashes throughout the tests. In terms of expressivity and ease of use, the system allowed for unrestricted movement but required some explanation to grasp it fully. One participant adapted quickly, while the other shifted from an exploratory first trial to a controlled second trial. This suggests a learning curve, but also potential depth - aligning with the notion that effective instrument design balances accessibility with long-term engagement. Addressing each clause of the framed research question:

- *Designing a breaking-specific sonification system* - Our mappings were aligned with the movement vocabulary specific to breaking, and sounds were influenced by classic breakbeats, reinforcing movement-sound cohesion.
- *Ensuring movement-aligned and responsive interaction* - Y-axis mappings were most effective, while stab-triggering and the vacuum effect require further iteration.
- *Reliability* - The system ran with high reliability, maintaining stable landmark tracking without crashes. However, improved Z-axis tracking would enrich the mappings.
- *Enhance expressivity without restricting movements* - The camera-based approach ensured free movement. While the system showed the potential to enhance expressivity in many ways, certain mappings, such as wrist-triggered samples, did not function properly, sometimes leading to user frustration.
- *Creating an intuitive system* - The system required some explanation before becoming intuitive. As with instrument design, a balance between accessibility and depth is essential; a system that is immediately understood may risk being too simple and the novelty would wear off. Ideally, it would embody "low entry fee with no ceiling on virtuosity" [28], although achieving this remains a challenge.

We hope to contribute to the broader discussion on how movement sonification can be meaningfully applied to culturally rich and physically demanding dance forms such as breaking. Through our project, we believe we have added valuable insights into how sonification can complement the dynamic and highly expressive movements in breaking. By addressing the need for responsive feedback and interactive systems that do not restrict the dancers range of movement, we have provided a foundation for more engaging and personalized sonification experiences in dance. Our work offers a proof of concept for how sound can be integrated into breaking in a way that reflects the dancers vocabulary, potentially opening up new doors for performers and extending the role of sound in live dance performances.

Design Guidelines: Vertical motion mapping should be prioritized, as changes in head height and body position are highly intuitive and align well with breaking's movements. Salient auditory feedback is essential to ensure that dancers perceive sonic changes clearly. Reducing unintended activations requires careful calibration of sensitivity thresholds, particularly for fast gestures like arm movements. Incorporating personalization and expanding the sonic palette, enhances engagement over longer performances. Optimize system responsiveness by minimizing latency and enable quick activation and release of effects to match the tempo of breaking. Integrate key breaking elements, such as spins and freezes, to reflect the full range of breaking's vocabulary. Basing mappings in cultural context by drawing inspiration from breaking's musical traditions, enhances authenticity and connection for the dancers.

Limitations and Future Work: The number of movement-sound mappings are relatively small, focusing mainly on vertical and lateral movements. Future works could explore additional mappings, such as spin-tracking or more nuanced footwork interaction. The study involved only two professional dancers, limiting the generalizability of the findings. A larger and more diverse group of participants, both amateurs and professionals, would provide a better understanding of how different dancers interact with the system. While the informal and exploratory nature of the evaluation allowed for rich qualitative feedback, a more structured and rigorous approach would strengthen the validity of our results. Future studies could also include more advanced technology, such as depth-sensing cameras or wearable sensors, to improve tracking accuracy and richer mappings. Finally, while the system was highly reliable, it could be improved in responsiveness and perceptual salience. Future iterations could focus on refining precisions of mappings, as well as making more subtle interactions, such as stereo panning, more perceptual salient to the dancer.

7. CONCLUSION

This study explored the movement sonification of the dance style *breaking* in an interactive system. Feedback from expert dancers suggests that a subset of the implemented movement-sound interactions were intuitive and functioned well, while others lacked responsiveness and were generally less satisfying to engage with. Overall, the participants found the system quite intuitive and easy to understand after some exploration, which also encouraged them to rethink their approach to the dance. Although at an early stage, this research provides an interesting foray into the real-time sonification of breaking, a style not yet widely explored within the context of dance sonification. Despite its clear limitations, our prototype serves as a useful proof of concept to demonstrate how movement and music can interact within the context of breaking. With the steadily increasing popularity of the dance form and growing

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ubiquity of real-time bodily interaction with digitally generated music, we believe that our research provides useful design insight that paves the way for more expressive performances and potentially novel directions for the evolution of the breaking genre.

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